

Insert new annex Annex 172A corresponding to Clause 172 as follows:

Annex 172A

(informative)

800GBASE-R PCS FEC codeword examples

This annex provides example PCS FEC codewords produced by the 64B/66B to 256B/257B transcoding, scrambling, pre-FEC distribution, and Reed-Solomon encoding defined in 172.2.4. This annex presents data in a tabular form. The contents of the tables are hexadecimal representations of the data and are transmitted to the symbol distribution function from left to right within each row starting from the top row and ending at the bottom row. The most significant bit of each hex symbol is transmitted first.

A constant transmission of idle control characters is sent to the 256B/257B transcoder in each flow of the 800GBASE-R PCS resulting in a continuous stream of 257-bit blocks per flow, $\text{tx_xcoded}\langle 0:256 \rangle$, shown in hex:

$\text{tx_xcoded}\langle 0:256 \rangle = 007000000000000000780000000000000078000000000000007800000000000000$

In this example, an alignment marker is due for insertion in each flow. The scrambler seed [see Equation (49–1)] just before the 257-bit block was scrambled is as follows, for each of the two flows:

For flow 0, the scrambler seed is $S\langle 0:57 \rangle = 24\text{E}6959\text{D}0\text{FA}5\text{DBD}$

For flow 1, the scrambler seed is $S\langle 0:57 \rangle = 1\text{FB}58857\text{D}81624\text{F}$

In each flow, after scrambling and inserting the alignment marker group am_mapped , the variable tx_scrambled_am is produced as shown in Table 172A–1 for flow 0 and Table 172A–2 for flow 1. In these tables, italicized characters are alignment markers and bold characters are padding for the alignment markers. In each flow, just prior to insertion, the seed of the PRBS9 generator used for padding is $P\langle 0:8 \rangle = 0\text{x}100$.

NOTE 1—A text file containing the patterns in Table 172A–1 and Table 172A–2 is available at <https://standards.ieee.org/downloads/802.3/>.

In each flow, after distributing into two message blocks and PCS FEC encoding, codewords c_A and c_B are produced. For each flow, c_A and c_B may be illustrated as two 5440-bit representations cx_A and cx_B as follows:

For all $i=0$ to 543:

$$\text{cx}_A \langle (10i+9):(10i) \rangle = c_A \langle i \rangle$$

$$\text{cx}_B \langle (10i+9):(10i) \rangle = c_B \langle i \rangle$$

The expanded codewords are shown in Table 172A–3 and Table 172A–4 for flow 0, and in Table 172A–5 and Table 172A–6 for flow 1 (bold characters are parity).

NOTE 2—A text file containing the patterns in Table 172A–3 through Table 172A–6 is available at <https://standards.ieee.org/downloads/802.3/>.

For both flows, the example codewords have $\text{tx_am_sf}\langle 2:0 \rangle = \{0,0,0\}$.

**Table 172A–1—Example tx_scrambled_am with alignment marker group
for 800GBASE-R PCS flow 0**

Index <i,j>	tx_scrambled_am<i,j>
<0:256>	0B2ACAB2ACAB2ACAB2ACAB2ACAB2ACAB2ACAB2ACA9324C9324C9324C9324C9324
<257:513>	192649926499264991B421184592144D2F445014D517435064290242CA669AA6A
<514:770>	1374DD314C5354D5354DD314C5334DD356D5B56D5B56D5B56D5B56D5B56D
<771:1027>	0B6ADAB66F9B96C9B76DDB96D1B76D1BF6E1B26CDBD6D9B8B7DA9F80678A8AD7B
<1028:1284>	1E4B2F2B80AF5D571B9B5D4CBDF1D00F9FF9714372729199DB35AFF4CDF2E9BF9
<1285:1541>	1D7E1AA0BD3D8BA16DAAB83C86F1C6A4C3D959E995921E601CD49D736213A81A3
<1542:1798>	028FD1912A2AD9283950E183FA3B0196372399DBC94932C982990B4102985E84A
<1799:2055>	15D1B04967A30D5178CE48F0569849A00846139561BD37228569FB24B7E4D4CC0
<2056:2312>	08CCD177010357E8DAB65B6F9160968075341AC958CFC828D8372DC66242646D8
<2313:2569>	026FEAA09736E915DD54C5F7E600ECFCBD7D7C207941C5D7182D8B95FE412ED12
<2570:2826>	014CCE446415F10CDC7BBAFB1CC4B442EF5737AFB4F3BED48D240C9095F5FD3A6
<2827:3083>	10219FBE135EDE0B73C15238C150007886562E30A50CB27DF341954DB8CA1C564
<3084:3340>	16598EAD3C229838CB9BEE7F793139EE1F89FDAD9071687CBEDF89B4E527A03CD
<3341:3597>	1C2044F260A85EE6EB506C25E7C1196298548B4B3576F1B084C83F2E3CB5FC5F6
<3598:3854>	1C9D40EF27471536FE9EB5E3BC3870D46755080EA6D29F09CF8FA697A738DD3EC
<3855:4111>	193FA8A7675CE03706A49019B93544E19DD66CE78E6A94BEBD07ECCAE7DF2076D
<4112:4368>	081919F234C00959E5AFFC9F83C909805378B5F3F2E4917F3E48B5DE4758CEA42
<4369:4625>	1D378C193CDD3D0F781ABC3529BF76A66B7C73A7230B517D96DAFF6A39F9EA9B6
<4626:4882>	05E852B6917E9D1410B6509E778629A516FB2BCEABA79F12241DBC94CDAFFFF02
<4883:5139>	1D98F5991744C9FC5393EPD6299558CB4F28DF3BF3C8636DBA505E3A296C7BD2E
<5140:5396>	00FAACFBFDBC1C29CED046C73C9AAAFE3DE8849ADAD16E864814846BB8CB889AD
<5397:5653>	04D11BFE0E8DEF5ECFDBE45D1EE4607B4B44DF874F8E976DD828FCFD38158A8FF
<5654:5910>	00EE28B635049BA9506024BA13862A1D6BAE22D0DB5DDB1F4DBE0F80E9EDBBCC6
<5911:6167>	1CD0769BC76446DA04136E7C6D19905DFB81AC3BFD931426901028D729E529F5A
<6168:6424>	14C3FF33243D7D2B5137B63359FD25B82CDE769D0F10D2E60A0386E0820BE8F4
<6425:6681>	1993F1FE2D6C60459F26A74BD0265C2959899E4C5B24196AFD51DB241CFCF9091
<6682:6938>	18EE61EA5A0D584DE8AC602618874BBA60A907115513BC960F6BE32C68F1F2458
<6939:7195>	042251FD6EF16662F5C99D9779B20B865B1401C96034C99550401BCB2792E5638
<7196:7452>	0CE3AE06476B769F33E5577CE4BA6D62071CABE3EA9561D6077FD23909568A25A
<7453:7709>	1BBC46AC3DF26A5BB0EA4FDBCBCFB42738A84007619C594E24AB396AFB5F05FBFF
<7710:7966>	076C96A407E106D494AA6B0C5168E1F335B812D7BC530CE8C6FC13F6C54EC2165
<7967:8223>	056CC06AFE2BD2A9894C4D1AD9E632F861208AD3894EFF0DE8B0291C486EA3281
<8224:8480>	094534A897031E6307E32C1906C85E99CD4696FC818D2B5EAF26A576DE149AE12
<8481:8737>	085AE9048EDDC5401827FAA937411FF35C10287236FFDC87E7E7E334B038EE3F9
<8738:8994>	014179E0E2359D3B961A1302FA4B7AC384F0564B15D7504DADBE3C65EE8F6F13A
<8995:9251>	08D40F4B092621CCA4919E81D0C15014204622D41814894DBE389C149E5E2257A
<9252:9508>	0E4772C6718DA5361ABFAAD60F5C32D304740DE671045CDAD6617125F4BBF457D
<9509:9765>	071A7D2B02B41F13971A22FE8A29F0A03C1CEC43CA0411D7872F18D12E0A2BD06
<9766:10022>	1344403844B360B4A61968D045941FF821511C2E95455C300EED811D314AD7018
<10023:10279>	0275A5C4A36F6B8B13AFAFFFCDFDBD9B13F00488494886CDEB13B31FC9F7941CF

**Table 172A–2—Example tx_scrambled_am with alignment marker group
for 800GBASE-R PCS flow 1**

Index <i,j>	tx_scrambled_am<i,j>
<0:256>	0B2ACAB2ACAB2ACAB2ACAB2ACAB2ACAB2ACA9324C9324C9324C9324
<257:513>	192649926499264991B421184592144D2F445014D517435064290242CA669AA6A
<514:770>	1374DD314C5354D5354DD314C5334DD356D5B56D5B56D5B56D5B56D5B56D
<771:1027>	0B6ADAB66F9B96C9B76DDB96D1B76D1BF6E1B26CDBD6D9B8B8196F7C588549274
<1028:1284>	063530AC615723489C7AA533420E2FF060068EBC8D8D6E6624CA500B320D16406
<1285:1541>	0181256F4EC1749EA25944C3B93E35583CE696199A91DE9020DB9EB3922FA719F
<1542:1798>	036FA98F2DAB3957C6AF1E7C05C4FE69C8DC662436B6CD367D66F4BEFD67A17B5
<1799:2055>	0A2E4FB6985CF2AE8731B70FA967B6500846139561BD37228569FB24B7E4D4CC0
<2056:2312>	1F86712D0B9C5715BC0B73ECCC6DD388DCC420940ACD6A761E9DBFD65F67A6F80
<2313:2569>	17D275B453FCD8EE9BBA94A522014E8FA3612BD59F1561BF74F4DFA4BBB186627
<2570:2826>	08F114445112D312B36734B261D20AC5BA8E888DFFF7AC7FBFDDCC270294A477B
<2827:3083>	129C411094B7035E81B92A2390C3A5F42EEB0FAFD8B4AB14E2F282A004EF203DF
<3084:3340>	1922EC8280625B274A3BE4165602BE1AA5D500E9B5E42CA8A1ABF23428487DCC0
<3341:3597>	05283BE31434B6768BA691A9E03AD081BDE41FD90DCF8E50CC9C6993D2632CE00
<3598:3854>	010D455AB1946A8AF1B27E7970F9C64099FE6DD0BF426CFDD965F0F62B29F497B
<3855:4111>	0A24DCC7CA2E026301236DF64D82DE1BA3C0782F57F00618A931EBD9CD53E5FDF
<4112:4368>	1FB0A3715E0D9FB90914C768F175C7CC935331B3C4575547B4C4C25B9ABCD8698
<4369:4625>	0C7BDEACF874C37E408742B8E1B1D1155793CD9A46DB62CED7FED0542B774E155
<4626:4882>	0FB5F72DEEF86145B1203BB9357A111B3DE41A6F68FF8EFBA05764C7C8A310219
<4883:5139>	14E0EFA4D7E1D11276142A97DC986CC8AD33952EB7415501160BC907D27957D78
<5140:5396>	1ADD66B6629E510BDC1C913AB02C7BD575C4165BA1F57D71C1C67C129EDCD0885
<5397:5653>	09357A0D4F3A2EDE5040F70E722B367A0B5995F07EDB2DA93118CA44E4A95BD8C
<5654:5910>	1F87871D3792E369DC0EE28B3700C7B21256A342A41541C21760FA2A817EB1718
<5911:6167>	1A08EF85796CA9198F49380C6834D413CD1E6AB22AAF3826244FF2FAE786966CE
<6168:6424>	0C17672A70D318BD56B86CAD4E6B5F8CF307FDEC8331ECC71F7918F94245C3F62
<6425:6681>	19F56BAEFA5211C1272E4F9D16CA2CD6F63E731C1F67495BB02209956EB212FF2
<6682:6938>	04CE02FCA580E394CD4BBEEE49A2724E89FCFF76F58F6A5C9554C369DAF03EA38
<6939:7195>	0D7A7417F605A52B76F10EB7D784A8C0D46CA4A46082E97C54682CCAD8128F4C8
<7196:7452>	1F062F513B79AA19647D271AEC2A27C172D10824306C524CFC5AD1A882EC269C47
<7453:7709>	16464DB070D86EA6F6F2DCC17BF64C0439C0DCC6F5E080B7FA5CF0BC17D4941FE
<7710:7966>	1E570C5B95F6A1EF2568FBA6A3E2AA3E10F92CFC848B7DDDC0C229DA995EF3238
<7967:8223>	0B019680216F92EFC7A77F2D846BF50FB5571C8B0597EDD4444C0D1ACD73ED0B2
<8224:8480>	0D4342D2956C0316F27A6CA376E43165DFF6E0BF72B3B4B684DD48B5C1E497305
<8481:8737>	19D3C8BC33383FBF0B145F73B019C77C25F7EF62FE3A30D7BF87ACDE21F33AB65
<8738:8994>	1B21A3FA20EB18693BA928B8E8B1481F9C1B4CAA136A317E5575FEE62626B4689
<8995:9251>	0217E9C2B11E1B9B07184C9B7188D658F7703770D3DB76533A2A6AD850A9C01B3
<9252:9508>	1407CB283808020B5602DA0A14AC8761931FDB8BE807E7D3D7A6ED35A656B92F8
<9509:9765>	059FC0379557A8410F5AA2B5D7F4A506BA87443FF05C4F26D9B1B7625AA0AAD8A
<9766:10022>	11C1338632F6EE337BA90C50DB4CDEC64C758D8B5FAF5A880C5C3C62FBCE1A7BD
<10023:10279>	0581091408463FFB4FD2C97DE7105B4108FC7FCF460728BDA613FD80FA8608928

Table 172A–3—RS(544,514) codeword A for 800GBASE-R PCS flow 0

Index $\langle i:j \rangle$	$cx_A \langle i:j \rangle$
<5439:5120>	A6A9AA6A9AA6A9AA6A9A64992649926499264992D886284BD280BA2609026519565D946559565196
<5119:4800>	6D5B56D5B56D5B56D5B57D936ED8B62D876CD9B6FB41F547DE7A6A07575BDEA014FDD8C499B97EE9
<4799:4480>	BCD56BC9A11D7D8256A6A999E2B1B315CA3131AA380C3B8A34339492684C810BD2EA92C331E3C619
<4479:4160>	401C8F61447E7EDCCA66077D56AFED4815C9AFE6D8B691236C7F7A4894AA7ECDCEBC10380639D209
<4159:3840>	48B9904D84BBDC645B57BEBEE96049F5CB27E7ACB40A8A0DE01AA509342C6CB8534AAEA22633BD93
<3839:3520>	3BFF204FC2FD8E502D0F4F285D750D83E51D13AB0DB043C68F7267B51CBF8F70ECC40896DC8CBE72
<3519:3200>	F95FC9B80795998ABB99CD56F5F37EFB6EC4E58D4BEB0FF21F679F490F9DD635A10C7B3FC2D5A52
<3199:2880>	56CFB4E62BB695B79D1BA97F4050533DD2C9BEAB1F1C1653FF6E99AC88A3FDE99B4FB1F3F6C82945
<2879:2560>	F7BE0EFB07B9C6CACB8F9096B308258EF228B01FF7AFCBA189D2C7E71EED3F207F163B58EEC0C02E
<2559:2240>	A8EEB169DD65CF86F319B82E36DAC8D8E09077DCA3009AC4A7569FFE11A96F735ED0F685E96028CE
<2239:1920>	E8265F8CDA9A39540A509936869A2A8267CE3ABC560BD80F08CBB0415749FACB149C42AFE68EBE9B
<1919:1600>	135A6E02C0AA5807906AE384CB74F99F6CB700FA5480D25D52D208FBC295570F642C546E2685255A
<1599:1280>	0FBFFD27F056F2A147C33B6F561BB1DF9B994E0351D95C8F9B7D104470FF3463815D28A958106069
<1279:960>	9B3324B302D6AC9DB6B2A10823B86055E0BACC50F6F847E434C7414E0DCD0D206DA50E4D75DB65E3
<959:640>	2EE470AD0DE1092E04A8812B44A3DB0E7A77A9DCC7194FEA0DA61C433F4435891EE97DBE750C8D16
<639:320>	51543380F0881388B114962019A6962D0D13FC4552E1D6EE5CD6A408EAF6237F5FECD980292CD9C8
<319:0>	4FC29D9798C2C55370750272E393736FF6FD38A3DF36A7A93002B47850A96CDE76854DA59062AD1B3

Table 172A–4—RS(544,514) codeword B for 800GBASE-R flow 0

Index $\langle i:j \rangle$	$cx_B \langle i:j \rangle$
<5439:5120>	A6A9AA6A9AA6A9AA6A9A64992649926499264992109A2C8A22ACAC250B426599565D946559765197
<5119:4800>	6D5B56D5B56D5B56D5B59DBB69DBB6FD936BD87695B98B52933A9D73B19571FCF51E72733ADF67EC
<4799:4480>	1FBD07455B4F238F0D7912780AE4841617E2A693857F8C049DEDD92A0C5A0CAA40D9798A89C6A164
<4479:4160>	621AAEC9A824EC9203458118BB4F4402C1635209D83314CA401576CEBA30193FFAE9EAE9B4FEB74
<4159:3840>	940447D476FAE46BA3D97965681352E5E1087DED3CC7100984C74C2FBFA631DA8335E5832CEDFCE4
<3839:3520>	3F1ADD1FE9B232567A2041A7782A7AA62A1348F6C874FEB76F8145CB65AF1ED61554AE7CA3EF4B63
<3519:3200>	999453AC3B093278719A71F4AE09D404C0A9F0014F09F279005A3A7FD289C4EB9B2C98F2C1EB0EFD
<3199:2880>	599CE431BEFD79C6542F4B572684F2A32D19EBCB2267BAD81FF1BA2F93274686A4F37361176C78DA
<2879:2560>	749AAB7CA1415E7EA8BD6B3B4026B13A65AF6317DEA7DDE1E0C8B2EBA5412FD1A83DA29055292039
<2559:2240>	5C08E6DB1B3EEE0EEECED94892079D98BEE35A6F6410294FE5866667D64598E5F680B711C33C3A08
<2239:1920>	E2FC9D1E0693FD2E999AC9C24FD4DBFC9C0338B490C6AC33A5A8111EE5E0A631E0D229F75199927B
<1919:1600>	1D0A36924C20B4F749870EAB79F355A4E35A72BE70FF713A8B76B5693CDDFC9FD73802E6072736DF
<1599:1280>	7EADC1281094A1A8B599A41478B9075190836BD54BCA458B19862D4729EC9418414E51D233C8FC4C
<1279:960>	F41597EE58EAF5250E435D7710ABE4B27F820C4EEFFE7B331C27F3D623B942CBE1AC1E46815EDB4C
<959:640>	B784B2F192531E6308A04606029071929114E71AB62C36A9D71958111ED9D0D7D45D8E353823A7F4
<639:320>	E53C08DE057AD8F1D0BD114438370C05B05422875160140E51C056BC5075F5CFFBDBF211044D7A33
<319:0>	EFBCE3BC2198756E3F23CE3783B4D1085B12943E924B87EF67FED64D92E66AE99B5E579C0F8B1BCB

Table 172A–5—RS(544,514) codeword A for 800GBASE-R PCS flow 1

Index $\langle i:j \rangle$	$cx_A \langle i:j \rangle$
<5439:5120>	A6A9AA6A9AA6A9AA6A9A64992649926499264992D886284BD280BA2609026519565D946559565196
<5119:4800>	6D5B56D5B56D5B56D5B57D936ED8B62D876CD9B6077EFA842E86550894AB215FEB02273B66468116
<4799:4480>	831A98355E22B271AA5959A5EDB273E5F63E326AC7F3C475CBCC6B6D97B37EF42D156D3CCE1C39E6
<4479:4160>	401C8F61447E7EDCC8C3859D481E6F5DA7648166C37FB37C3EE4B28719774A5CA6C6AF6A8BB5FA37
<4159:3840>	E4544116469CC326A38AF61AF7FB21A50EF20AD25E949C26FA35DBF354F40144F4FC262325C13C0D
<3839:3520>	AC015D6EA6EAB85DF128C7F4BD15625C360F80EC0A18E4BDCD612ADAC6C73CDCF321DB85DDF1F66A
<3519:3200>	D27244FA401226FED3C5D040F547D72AFBF284F5EFE5178BC75679B5765BD227A0C9EFE1DFB2E21C
<3199:2880>	88B272591BFE8A82EFAA9DCFB46E0D5AC404CB7EE3507C446CB25FE1EE45519344E9CBB40AF40BE
<2879:2560>	FAB6B3694F1DF911A1D5341F0C77CCBAC21AC8AC7448270F35D07EADB725531491BCF0D9763E0668
<2559:2240>	C612485954D0A2FFD0C7DC69E6249258C2BE2C4D73E247D1619CDCD5857AB0CE58FFC109A6FF7D3E
<2239:1920>	1D2685D529483C9369ACE7FE0A91106A921645FAE0199EFA2C2E7F935DD2656DCAF9EBFA25ADBAD7
<1919:1600>	486B0494108FB4106F2F307912AFC4AC7C85A4485187EAC46E11C980E15D4F7A0324E7630415FF43
<1599:1280>	2BDFE61FD4EF38B8AE2A3E09F7D981B73BD43969DA3F7FBA21C28EA837B7C8B58DF1645A06D7B66C
<1279:960>	3B74D3B69D695761D5D2B9CF4C1F43EFA6043FDFC7FAC5E211D5C268BCC6BB8E8519CFA66B64FFFD
<959:640>	8C8B1FA0D47618EB66B1BBCEEB6DCCD59503627C0E208A010A70B19E8CFCF5D96B503E03EAA82156
<639:320>	EB829E17FFF236646D41150E6DEB666116DC6CD7F58AD747A3B0DA5A2462CB69323882F8C62F47C8
<319:0>	7C1101E2E9D7DE149660AA91354E2A473CF6434EB00C349782F3FF0E85F3DDEDEF69F4DBED4633BD

Table 172A–6—RS(544,514) codeword B for 800GBASE-R PCS flow 1

Index $\langle i:j \rangle$	$cx_B \langle i:j \rangle$
<5439:5120>	A6A9AA6A9AA6A9AA6A9A64992649926499264992109A2C8A22ACAC250B426599565D946559765197
<5119:4800>	6D5B56D5B56D5B56D5B59DBB69DBB6FD936BD8766986849163C6A27C72658E030AE18D8CC5209813
<4799:4480>	2072F4B9A470EC7CF186E24405E744E62BEDA5537A8073FB621226D5F3A5F355BF26867576395E9B
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